

FORM 2
THE PATENT ACT 1970
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&
The Patents Rules, 2003
COMPLETE SPECIFICATION
(See section 10 and M.13)

AUTOMATED ACCIDENT DETECTION AND TRACKING SYSTEM USING IOT

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The following specification particularly describes the invention and the manner in which it is to be performed.

FIELD OF INVENTION

This invention relates to Accident Detection and Tracking system

BACKGROUND OF INVENTION

In winter the situation changes rapidly. Due to heavy work, vehicles such as cars and motorcycles are increasingly used. People become victims of excessive speed while driving. Many times, victims cannot get help in a timely manner because no one knows their situation. The most important modern influence for the Accident is the use of technology, such as changing music channels, Drivers often get traffic information from mobile phones, GPS and other devices. Drivers may be distracted by other road users, eating, drinking coffee or other beverages. Automakers such as General Motors and BMW have adopted automatic collision warning systems. They use sensors in the vehicle, such as accelerometers and airbag sensors. Unfortunately, most cars do not have automatic collision warning. Therefore, it was replaced by smartphones that not only capture the situation but also send this information to the authorities. Using mobile phones as an alternative to traffic accidents. This alert has many benefits, including mobility and the ability to provide collision warning for all types of vehicles, including bicycles.

SUMMARY OF INVENTION

The main purpose of the invention is to help and save the life of people who had met with an accident and struggling for their life. When an incident occurs, notifications are sent to registered mobile phones. The application will continue to use the MCU node and GPS module to check if an accident has occurred and report the incident to the mobile phone, email address and nearby medical center. The system can track and display the location of the site. It is known that urban traffic is increasing day by day and has become a big problem. This is because vehicles such as cars, buses, trucks and bicycles are used more frequently and cause more accidents. This is also because drivers like to drive too fast, which is very dangerous. There is a special application that notifies family members, ambulance and the location of the victim in case of an accident. This will help find and manage help quickly. Sensors such as vibration sensors, alcohol sensors, heartbeat sensors, temperature sensors are used here. GPS is used to track the location and notify the relevant number via SMS. The MCU node displays the vehicle's location in latitude and longitude. But finding it requires a lot of human effort. If the vehicle breaks the virtual fence or exhibits excessive alcohol content, the emergency notification system will send an instant notification to the remote control database.

Detailed Description of the Invention

The systems(200) are made up of many sensors such as an accelerometer sensor(201) Arduino board(104),SIM GSM(106),Data server(107),GPS(108),Gyro sensor(103), which are installed in the vehicle to collect information about the vehicle's location and direction at the time of the accident. Use accelerometers and gyroscopes to measure dynamic parameters such as vibration, shock, and high-speed compression or tension. The system uses biosensors, vibration sensors (102), and airbag sensors to measure crash frequency.

GPS/GPRS (108) modem is used to locate the accident location. The car (100) also has a key that can be started in emergency situations such as a heart attack. The collision detection module uses GSM (106) and GPS (108) technology. Tactile devices in the safety space monitor the operation of the driver's seat and seat belt. When an event occurs, the accelerometer sensor will detect the vibrate violently using MEMS (203) and the rotating rotors to measure the orientation and movement of an object, GNSS Network(204) gives the live location information to the Arduino board and the GSM module(205) will receive important messages. Identifying the victim is also easy because this process covers all smartphone users. In the event of an accident, sensors in the smartphone determine the vehicle's GPS location (107), speed and voice signature and send this information to a central server via cellular connection, which is then sent to the emergency service and emergency contact numbers (206). This system is a prototype of collision warning and vehicle tracking using GSM and GPS modems and Raspberry Pi.

DETAIL DESCRIPTION OF DRAWINGS

Fig 1 is a Structural diagram of an Accident detection and tracking system.

Fig 2 is a schematic flow chart of an Accident detection method according to an embodiment of the present invention;

Step 1(200) depicts the beginning of the system which boots up the system

Step 2(201) is where the sensor measures the acceralation of a car in the time of accident

Step 3(202) is if the acceralation is less than 60Hz the system detects the accident and sends the input to Arduino (104)and if it is greater than 60Hz the loop will restart to the beginning of the system.

Step 4(203) The MEMS and rotating rotors in the system will measure the orientation and movement of an object.

Step 5(204) GNSS Network gives the information of a live location of a vehicle to the data server.

Step 6(205) The GSM modem (105) Swill send the information to centralized data server.

Step 7(206) this data server (106) will receive the location of a vehicle and forward it to the contact number saved in the data server.

Step 8(207) the program can be closed and this will end all its functionalities in the system.

We Claim

1. An Accident detection and Tracking system (101) comprising an Accelerometer sensor (102) which is used to detect the orientation of a vehicle and motion changes when an accident occurs and also measures the vehicle's acceleration and sends the data to an Arduino board (104);
2. An Accident detection and Tracking system (101) as claimed in claim1 wherein the Gyro sensor (103) uses a micro electro mechanical systems -MEMS (203) and rotating rotors with three axes to measure the device's orientation and acceleration of a vehicle based on the displacement of mass and send the information to GPS (107);
3. An Accident detection and Tracking system (101) as claimed in claim2 wherein a GPS tracking system (107) uses the Global Navigation Satellite System (204) network that incorporates a range of satellites that use microwave signals transmitted to GPS(107) devices to share information about the live location of a vehicle to GSM (105);
4. An Accident detection and Tracking system (101) as claimed in claim3 wherein the GSM (105) requires a SIM card to register a connection with the service providers in the nearby location, when the accident occurs the live location of a vehicle determined by GNSS (204)is then sent to emergency contacts saved in data server (205) via GSM(105);



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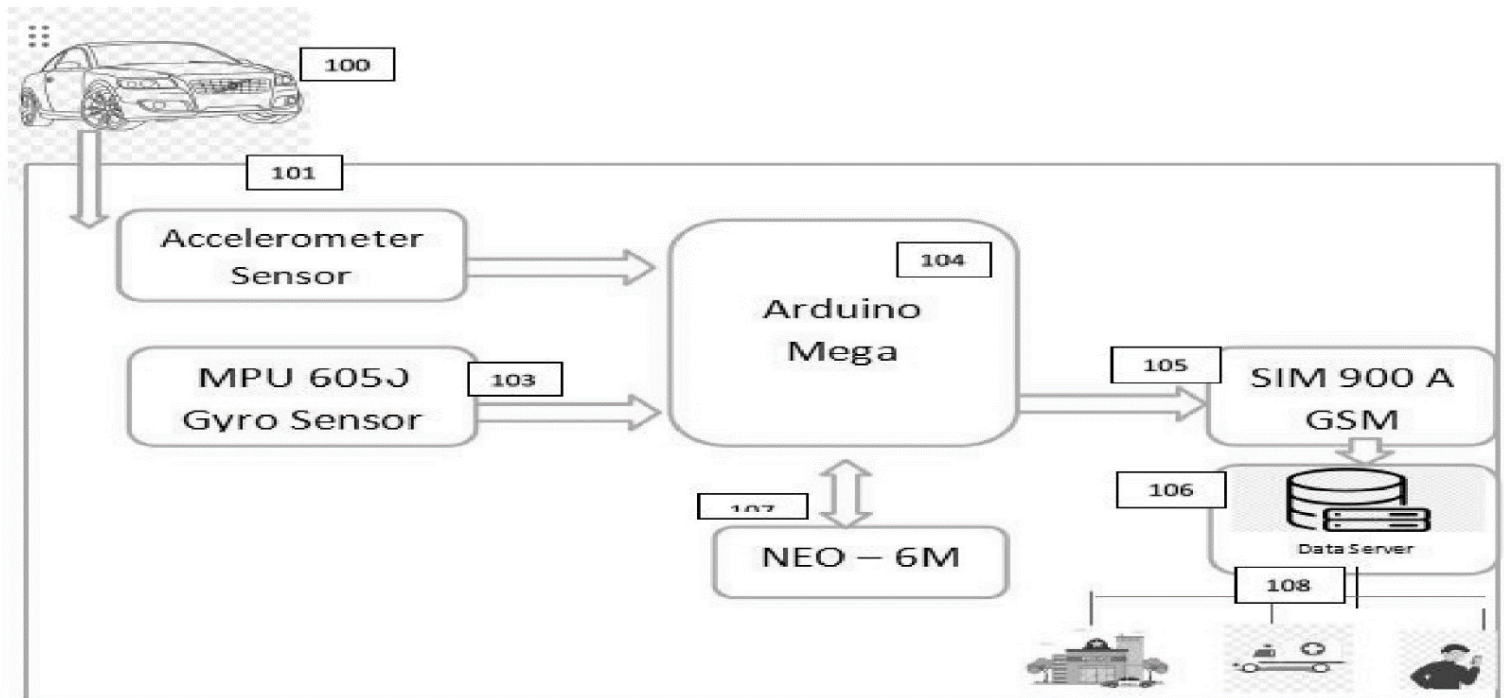
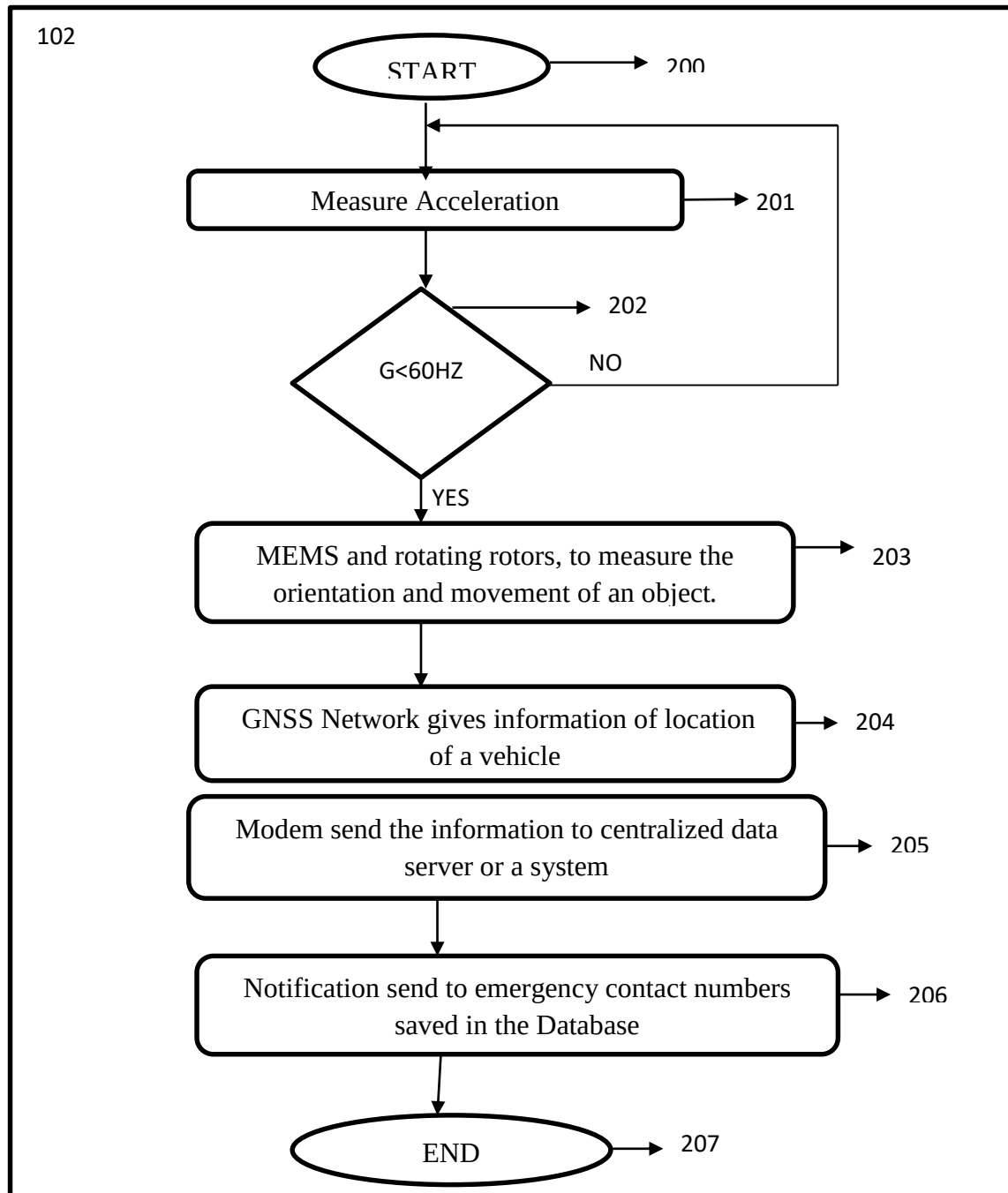


FIGURE 1

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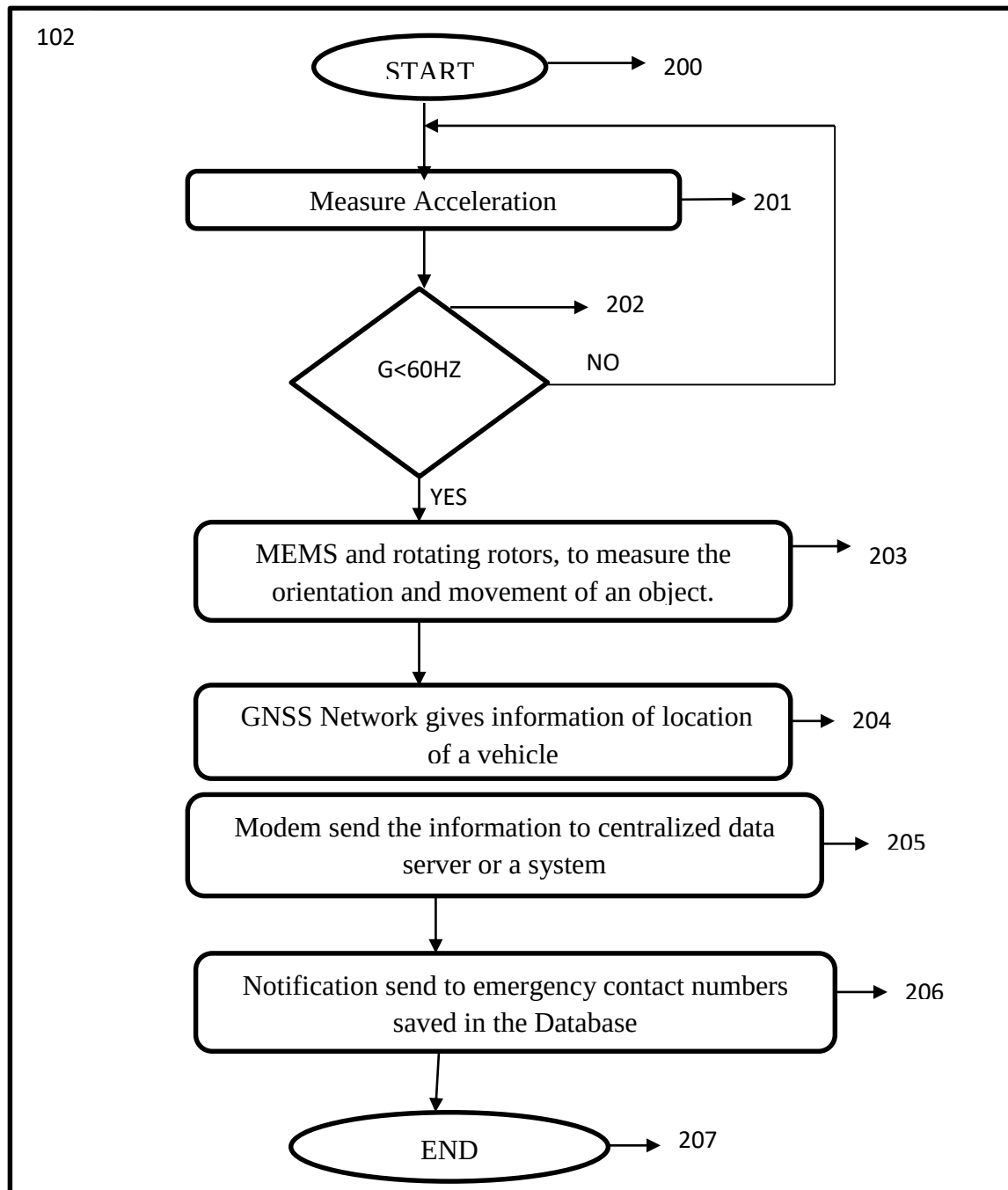


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ABSTRACT

An Accident detection and Tracking system (101) consist of an Accelerometer sensor (102) which is used to detect the orientation of a vehicle and motion changes when an accident occurs and also measures the vehicle's acceleration and sends the data to an Arduino board (104);



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